# Question sheet A answers.

Q1) a) Differentiate wrt $t$.

i) $y = t^{10}\sin(2t)$

Use product rule: $f = gh \Rightarrow f' = g'h + gh'$

$y' = 10t^9\sin(2t) + 2t^{10}\cos(2t)$

$$\boxed{= 2t^9(5\sin(2t) + t\cos(2t))}$$

ii) $y = e^{3t^7}$. Use chain rule: $f(t) = g(h(t))$

$f'(t) = h'(t) \cdot \frac{dg}{dh}$

$$\boxed{y' = 21t^6 e^{3t^7}}$$

iii) $y = t^2 e^{5t}$ (Prod + chain rule)

$y' = 2te^{5t} + t^2 \cdot 5e^{t5}$

$$\boxed{= te^{5t}(2 + 5t)}$$

iv) $y = (2t-3)/\sqrt{t}$

$\frac{1}{\sqrt{t}} = \frac{1}{t^{1/2}} = t^{-1/2}$

$\therefore y = (2t-3) * t^{-1/2} = 2t^{1/2} - 3t^{-1/2}$

$$\boxed{y' = t^{-1/2} + \tfrac{3}{2}t^{-3/2}}$$

(v) $y = (2t^2 + \sin t + 2)^3$ Chain rule

$$\boxed{y' = 3(2t^2 + \sin t + 2)^2 \cdot (4t + \cos t)}$$

(vi) $y = \dfrac{\cos(2t) + 1}{t^2}$ Product rule

$$y = (\cos(2t) + 1) \cdot t^{-2}$$

$$y' = -2\sin(2t)t^{-2} + -2t^{-3} \cdot (\cos(2t) + 1)$$

$$= \frac{-2\sin(2t)}{t^2} - \frac{2(\cos(2t) + 1)}{t^{-3}}$$

$$\boxed{= \frac{-2}{t^3}\left(t\sin(2t) + \cos(2t) + 1\right)}$$

vii) $y = t^{2t}$ * Slightly difficult *

Take logarithms

$$\ln(y) = \ln(t^{2t}) = 2t\ln(t)$$

Use <u>implicit differentiation</u>

$$\frac{d}{dt}\ln(y) = \underbrace{\frac{d}{dt} 2t\ln(t)}_{\text{Product rule}}$$

$$\frac{d}{dt}\ln(y) = 2\ln(t) + 2 = 2(\ln(t) + 1)$$

Solve for $\dfrac{dy}{dt}$

$$\frac{d}{dt}\ln(y) = \frac{dy}{dt}\cdot\frac{1}{y} \qquad \text{Why?}$$

Chain rule: $\frac{du}{dx} = \frac{du}{dy}\cdot\frac{dy}{dx}$

Let $u = \ln(y)$, $x = t$

$$\frac{d\ln(y)}{dt} = \frac{d}{dy}\ln(y)\cdot\frac{dy}{dt}$$

$$= \frac{1}{y}\frac{dy}{dt}$$

$$\frac{1}{y}\frac{dy}{dt} = 2(\ln(t)+1)$$

$$\frac{dy}{dt} = 2(\ln(t)+1)\cdot y$$ ← Substitute original expression for $y$

$$\boxed{\frac{dy}{dt} = 2(\ln(t)+1)\cdot t^{2t}}$$

viii) $y = e^{e^{10t}}$ $\qquad (y = \exp(\exp(10^t)))$

Chain rule twice

$y = e^{e^{10t}} = f(g(h(t)))$

$h(t) = 10^t$

$g(h) = e^h$

$f(g) = e^g$

Chain rule on $f$:

$$f' = g'\cdot\frac{df}{dg}$$

Chain rule on $g$

$$g' = h' \frac{dg}{dh}$$

Substitute into $f'$:

$$y' = f' = h' \cdot \frac{dg}{dh} \cdot \frac{df}{dg}$$

$$= 10 \cdot \frac{d}{d(10t)} e^{10t} \cdot \frac{d}{de^{10t}} e^{e^{10t}}$$

$$= 10e^{10t} \cdot e^{e^{10t}}$$

$$= 10\, e^{10t + e^{10t}}$$

$$= \boxed{10 \exp(10t + e^{10t})}$$

ix) $y = \dfrac{t^2+1}{t+1}$ Quotient rule: $\left(\dfrac{f}{g}\right)' = \dfrac{f'g - fg'}{g^2}$

$$y' = \frac{2t(t+1) - (t^2+1)}{(t+1)^2} = \frac{2t^2 + 2t - t^2 + 1}{(t+1)^2}$$

$$= \boxed{\frac{t^2 + 2t + 1}{(t+1)^2}}$$

x) $y = t^5 \ln(t) + \cos t$

$$y' = 5t^4 \ln(t) + t^4 - \sin(t)$$

$$= \boxed{t^4(5\ln(t) + 1) - \sin(t)}$$

① (b) Integrate the following:

i) $\int t^{10}\,dt = \boxed{\frac{1}{11}t^{11} + C}$

ii) $\int \frac{t^{12}+t^5}{t^{10}}\,dt = \int t^2\,dt + \int t^{-5}\,dt = \boxed{\frac{1}{3}t^3 + \frac{1}{-4}t^{-4} + C}$

iii) $\int t^{-8}\,dt = \boxed{\frac{1}{-7}t^{-7} + C}$

iv) $\int \frac{1}{2t}\,dt = \frac{1}{2}\int t^{-1}\,dt = \boxed{\frac{1}{2}\cdot\ln(|t|) + C}$

v) $\int \frac{1}{4t+3}\,dt$ Let $u = 4t+3$. $\frac{du}{dt} = 4 \Rightarrow dt = \frac{du}{4}$

Substitute for $dt$:

$\int \frac{1}{u}\cdot\frac{1}{4}\,du = \frac{1}{4}\ln(u) + C = \boxed{\frac{1}{4}\ln(4t+3) + C}$

(Or memorise this standard integral)

vi) $\int \frac{1}{t(t-1)}\,dt$ Separate this integrand using partial fractions

We want: $\frac{1}{t(t-1)} = \frac{A}{t} + \frac{B}{(t-1)}$

Cross multiply by $t(t-1)$

$$1 = A(t-1) + Bt = At - A + Bt$$

Now compare coefficients by collecting $t$

$$0t + 1 = t(A+B) - A$$

$$A + B = 0$$
$$A = -1$$
$$\therefore B = +1$$

$$\therefore \int \frac{1}{t(t-1)}\,dt = \int \frac{-1}{t}\,dt + \int \frac{1}{t-1}\,dt$$

$$= -\ln(|t|) + \ln(|t-1|) + C$$

$$= \boxed{\ln\left|\frac{t-1}{t}\right| + C}$$

vii) $\int e^{10t}\,dt$ Let $u = 10t \Rightarrow \frac{du}{dt} = 10 \Rightarrow dt = \frac{du}{10}$

$$= \frac{1}{10}\int e^{u}\,du = \frac{1}{10}e^{u} + c = \boxed{\frac{1}{10}e^{10t} + c}$$

viii) $\int (3t+10)^3\,dt$ $u = 3t + 10$ $\frac{du}{dt} = 3$

$$= \frac{1}{3}\int u^3\,dt = \frac{1}{12}u^4\,dt = \boxed{\frac{1}{12}(3t+10)^4}$$

ix) $\int \cos(5t+3)\,dt \qquad u = 5t+3 \qquad \frac{du}{dt} = 5$

$= \frac{1}{5}\int \cos(u)\,dt = \frac{1}{5}\sin u + C = \boxed{\frac{1}{5}\sin(5t+3) + C}$

x) $\int e^{2t+1}\,dt \qquad u = 2t+1 \qquad \frac{du}{dt} = 2$

$= \frac{1}{2}\int e^{u}\,dt = \frac{1}{2}e^{u} + C = \boxed{\frac{1}{2}e^{2t+1} + C}$

① c) Curve sketching

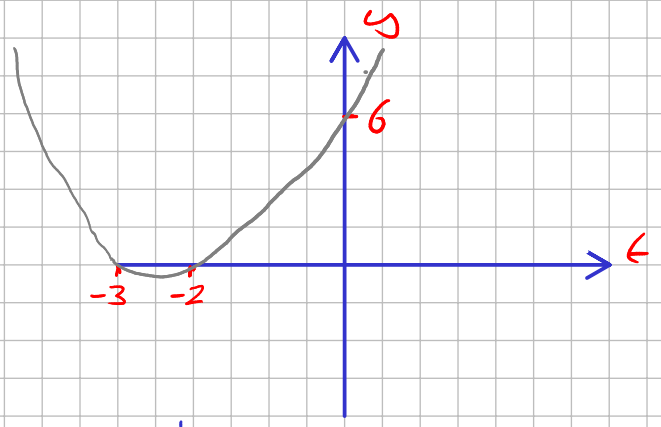
i) $y = t^2 + 5t + 6$

Key points:

@ $y=0$, $t=-3$, $t=-2$

@ $t=0$, $y=6$

We know this is a +ve parabola due to the leading $t^2$ term. Plugging in the key points.




ii) $y = \frac{1}{t+2}$

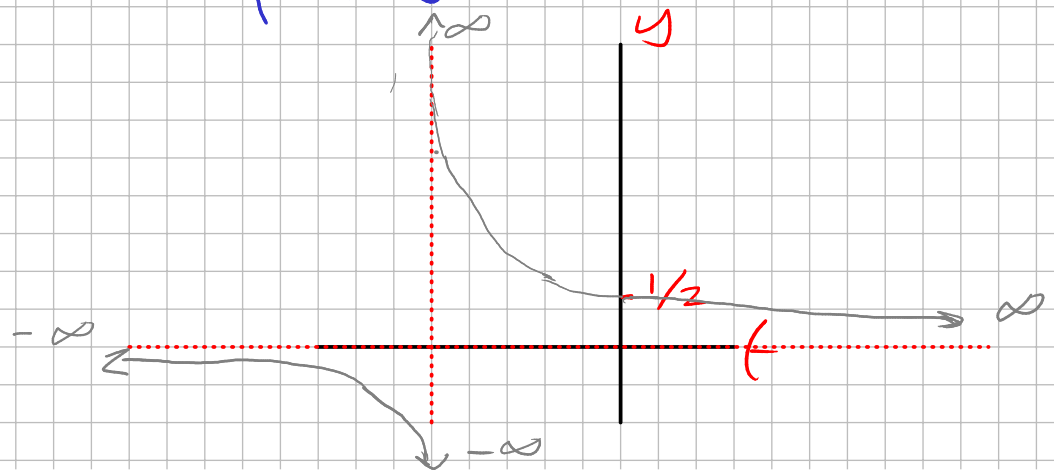
@ $t=0$ $y=\frac{1}{2}$

@ $t \to 2^{\oplus}$, $y \to +\infty$

@ $t \to 2^{\ominus}$, $y \to -\infty$

@ $y \to 0^{\oplus}$, $t \to +\infty$

@ $y \to 0^{\ominus}$, $t \to -\infty$



iii) $y = 2te^t$ | @ $y=0$, $t=0$ @ $t \to -\infty$ $y \to 0$
@ $t \to \infty$, $y \to \infty$

$$y' = 2e^t + 2te^t = 2e^t(1+t)$$

$$0 = 2e^t(1+t)$$

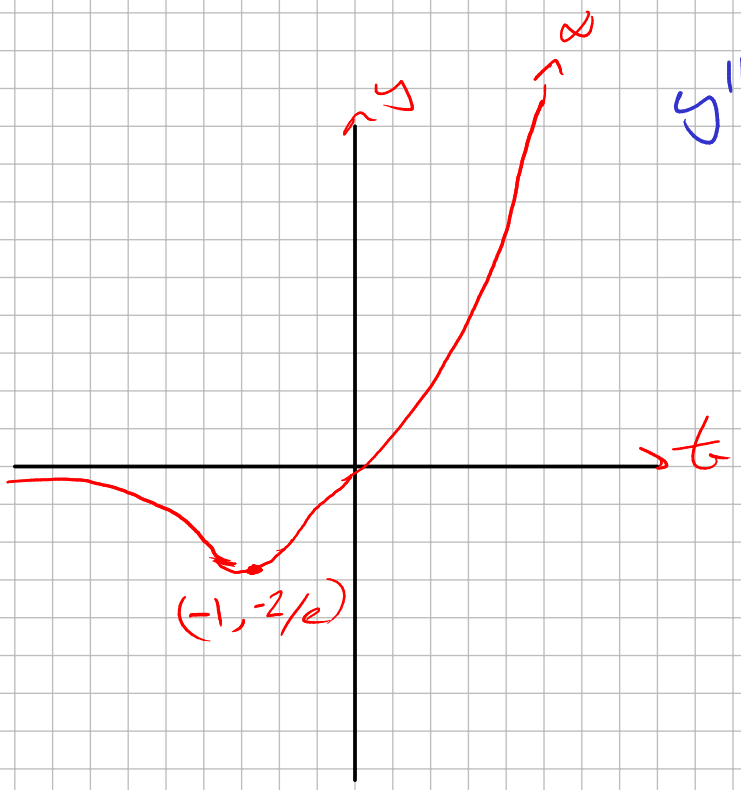
Turning point @ $t = -1$

2nd derivative $\Rightarrow$ Min or max?

$$y'' = 2e^t + 2e^t + 2te^t$$
$$= 4e^t + 2te^t$$
$$= 2e^t(2+t)$$

$$y''(-1) = 2e^{-1}(2-1) > 0$$

$\therefore$ Local minimum @ $(-1, -2e^{-1})$

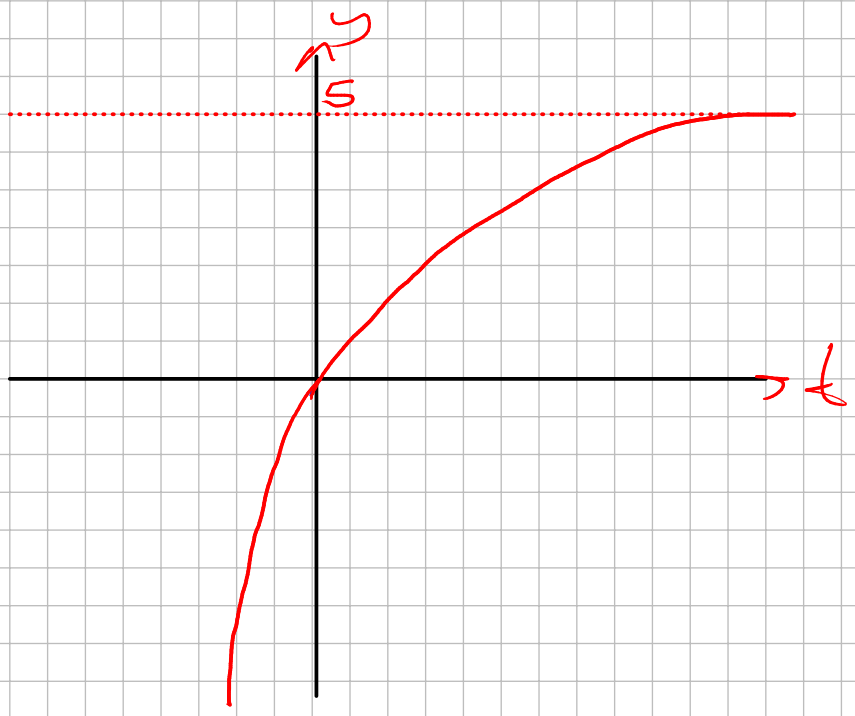


iv) $y = 5(1 - e^{-10t})$

$= 5 - 5e^{-10t}$

@ $t=0$, $y=0$ @ $t \to -\infty$ $y \to -\infty$

@ $t \to \infty$ $y \to 5$

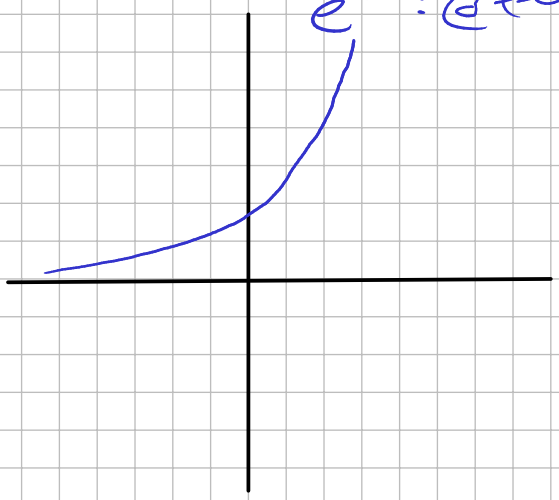



v) $y = e^{t/4}\cos(6\pi t)$

Split this up:

$\cos(6\pi t)$: Period $= \frac{2\pi}{6\pi} = 1/3$

$e^{t/4}$: @ $t=0$, $y=1$



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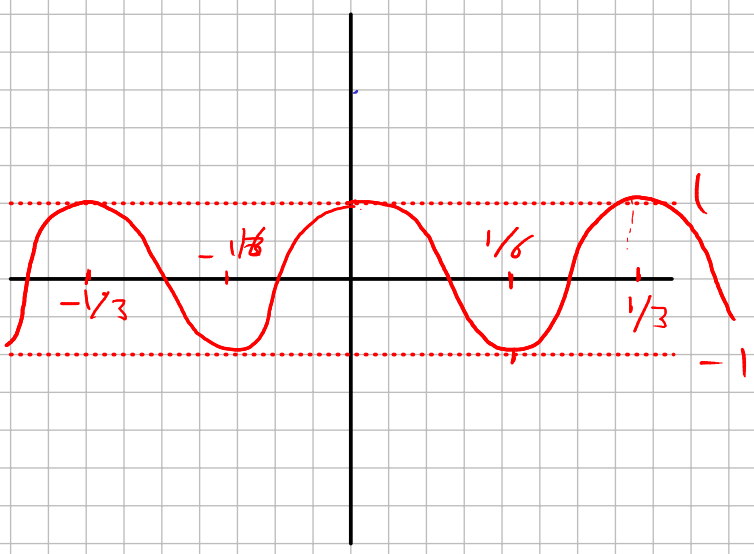
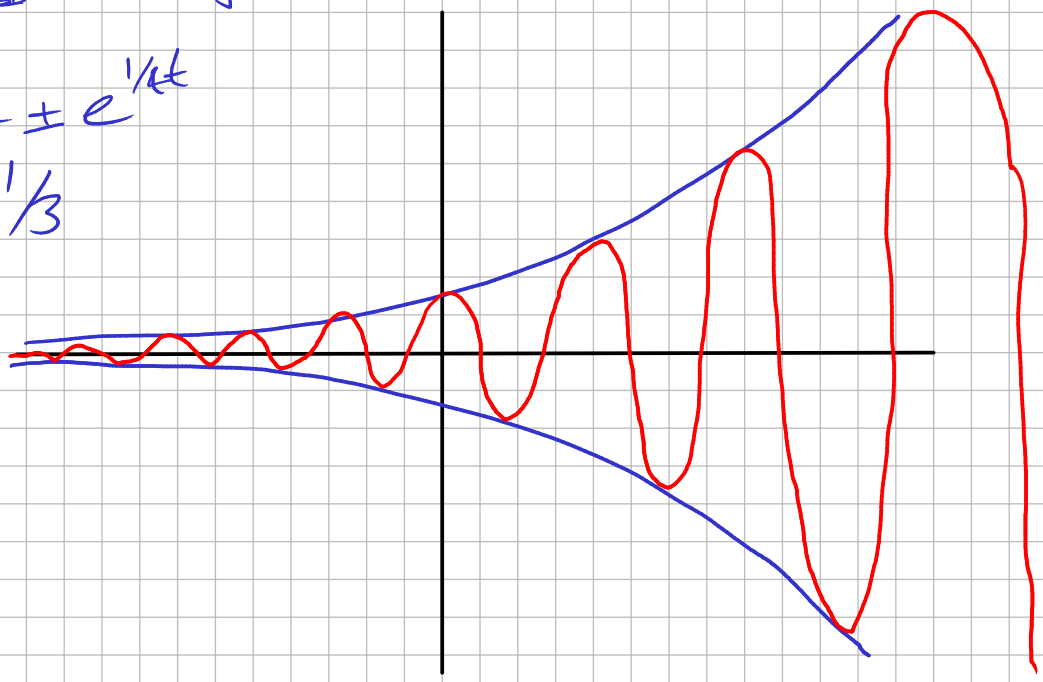


Multiplying these functions gives us a sinusoid bounded by an envelope

Envelope $= \pm e^{1/4t}$

Period $= 1/3$



(B d) Solve for $t$

i) $2^t = 32 \Rightarrow t\ln(2) = \ln(32)$

$$\Rightarrow t = \frac{\ln(32)}{\ln(2)} = 5$$

ii) $2^{3t} = 15 \Rightarrow 3t\ln(2) = \ln(15)$

$$t = \frac{1}{3}\frac{\ln(15)}{\ln(2)} = \frac{\ln(15)}{\ln(2^3)} = \frac{\ln(15)}{\ln(8)}$$

iii) $2^t = 4^{3t} \Rightarrow t\ln(2) = 3t\ln(4)$

Only $t=0$ solves this?

iv) $5^{t-1} = 4^{1-3t} \Rightarrow (t-1)\ln(5) = (1-3t)\ln(4)$

$$t-1=\frac{\ln 4}{\ln 5}-3t\frac{\ln 4}{\ln 5}$$

$$\Rightarrow t-1+3t\frac{\ln 4}{\ln 5}=\frac{\ln 4}{\ln 5}$$

$$\Rightarrow t\left(1+3\frac{\ln 4}{\ln 5}\right)-1=\frac{\ln 4}{\ln 5}$$

$$\Rightarrow t\left(1+\frac{\ln 64}{\ln 5}\right)=\frac{\ln 4}{\ln 5}+1$$

$$t=\frac{\frac{\ln 4}{\ln 5}+1}{\frac{\ln 64}{\ln 5}+1}$$

$$=\frac{\frac{\ln 4}{\ln 5}+\frac{\ln 5}{\ln 5}}{\frac{\ln 64}{\ln 5}+\frac{\ln 5}{\ln 5}}=\frac{\frac{\ln(20)}{\ln(5)}}{\frac{\ln(320)}{\ln(5)}}=\frac{\ln(20)}{\ln(320)}$$